

Panel 1: Data Collection, Preprocessing & Embedding

Steps 1-2: Raw Data

dog (2s), cat (4s), lion (12s),
tiger (12.5s), mouse (20s), rat (30s)

Step 4: Word Embeddings (spaCy)

dog: [0.12, -0.45, 0.78, ..., 0.23]
cat: [0.15, -0.42, 0.81, ..., 0.25]
300-dimensional vectors

Step 3: Preprocessing

Text normalization
Lowercase conversion
Punctuation removal
Stop word filtering

Step 5: Vector Validation

Filter valid embeddings
Check vector norms
Remove OOV words